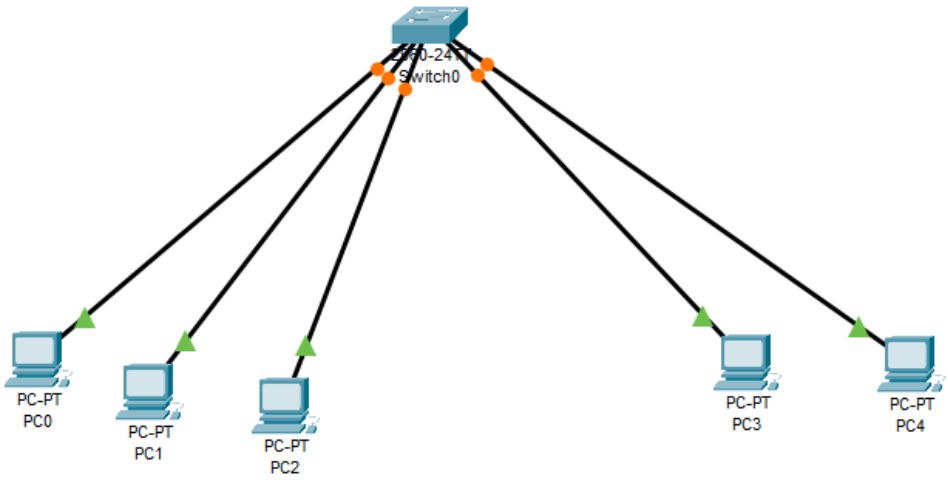
Day 3: Switch and Router Configuration Basics Theory :Intro to VLANs and static routing. Practical :Create and configure VLANs on a switch. Task: Configure a static route between two routers in a simulation.

Tasks:-

1. Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

Ans.:-



**2. Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.

Hint: Use the interface range command to select multiple ports efficiently.**

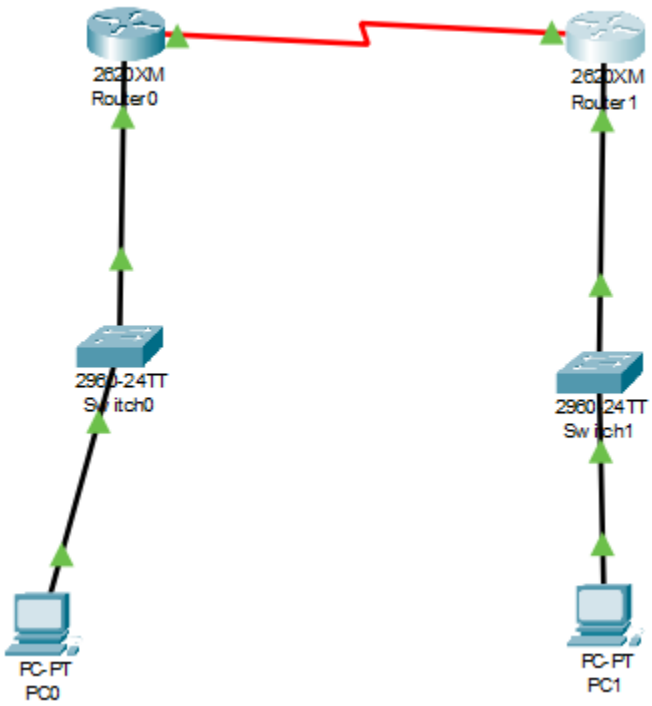
Ans.:-

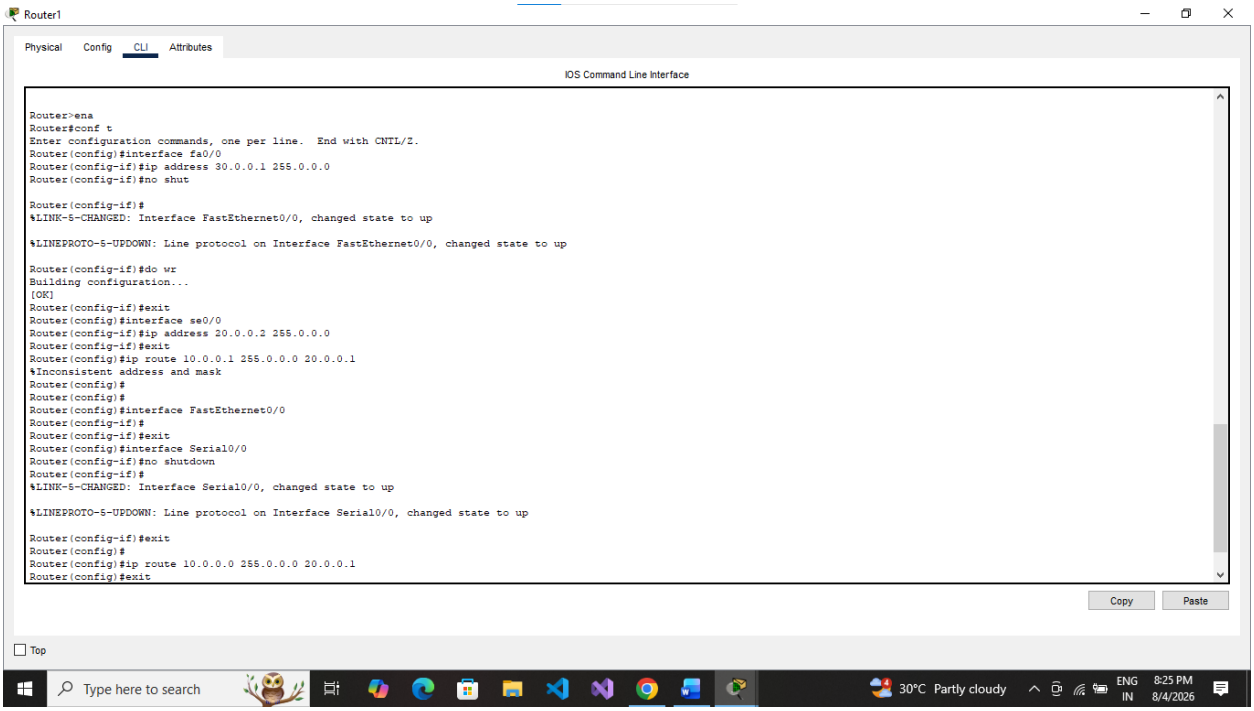
```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   sales                            active    Fa0/1, Fa0/2, Fa0/3
20   support                          active    Fa0/4, Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

3. Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

Ans.:-





**4. Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.

Hint: Think about router-on-a-stick or Layer 3 switch concepts.**

Ans.:-

Switch> enable

Switch# conf t

Switch(config)# vlan 10

Switch(config-vlan)# name Sales

Switch(config-vlan)# exit

```
Switch(config)# vlan 20
Switch(config-vlan)# name Support
Switch(config-vlan)# exit

Switch(config)# interface fastEthernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit

Switch(config)# interface fastEthernet 0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit

Switch(config)# interface vlan 10
Switch(config-if)# ip address 192.168.10.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# exit

Switch(config)# interface vlan 20
Switch(config-if)# ip address 192.168.20.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# exit
```

Why the Ping Fails:-

- Sales (VLAN 10) and Support (VLAN 20) are separate broadcast domains.
- A Layer 2 switch forwards traffic only within the same VLAN.
- Since no router or Layer 3 switch is configured, devices in different VLANs cannot communicate.
- Therefore, the ping request from VLAN 10 never reaches the device in VLAN 20.

How to Allow Inter-VLAN Communication:-

To enable communication between the two VLANs, configure Inter-VLAN Routing using one of these methods:

1. **Router-on-a-Stick** :-
    - Connect the switch to a router using a trunk link.
    - Create router subinterfaces (e.g., G0/0.10 and G0/0.20) with IP addresses for each VLAN.
    - Configure the switch port connected to the router as a trunk.
2. **Layer 3 Switch** :-
    - Create VLAN interfaces (SVIs) for VLAN 10 and VLAN 20.
    - Assign IP addresses to the SVIs.
    - Enable routing with the ip routing command.

After configuring either method, devices in the Sales and Support VLANs will be able to ping each other successfully.